

Report on

GAN-based Image De-noising and Multi-Angle View Generation

Submitted By

Md. Khairul Islam	ID: 22231203041
Md. Aman Ullah	ID: 22231203030
Shabbir Bin Zia	ID: 22231203018
Md. Zehad Khan	ID: 22231203004
Waliullah Khan Roky	ID: 22231203026

Submitted To

Sondip Poul Singh

Assistant Professor, Dept. of CSE, BUBT

Md. Shahiduzzaman

Assistant Professor, Dept. of CSE, BUBT



Department of Computer Science and Engineering

Bangladesh University of Business and Technology

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Declaration

We hereby declare that the project work entitled “GAN-based Image De-noising and Multi-Angle View Generation” has been carried out by us under the supervision of our respective project supervisor. This report is the outcome of our own study, analysis, and implementation. To the best of our knowledge, no part of this work has been submitted to any institution or organization for the purpose of obtaining any degree, diploma, or certificate. Any materials, ideas, or data taken from external sources have been properly cited and acknowledged throughout this report.

Signatures of Authors

Md. Khairul Islam

ID: 22231203041

Md. Aman Ullah

ID: 22231203030

Shabbir Bin Zia

ID: 22231203018

Md. Zehad Khan

ID: 22231203004

Waliullah Khan Roky

ID: 22231203026

Approval

I hereby certify that the research work presented in this report, entitled “GAN-based Image De-noising and Multi-Angle View Generation”, has been carried out by Md. Khairul Islam, Md. Aman Ullah, Shabbir Bin Zia, Md. Zehad Khan, and Waliullah Khan Roky under my supervision. To the best of my knowledge, the work submitted in this report is original and has not been submitted to any other institution for the award of any degree, diploma, or certification. I further certify that the report fulfills the academic requirements and standards of the Bachelor of Science in Computer Science and Engineering.

Supervisor

Sondip Poul Singh

Assistant Professor

Department of Computer Science & Engineering

Bangladesh University of Business and Technology

Dhaka, Bangladesh

Course Teacher

Md. Shahiduzzaman

Assistant Professor

Department of Computer Science & Engineering

Bangladesh University of Business and Technology

Dhaka, Bangladesh

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Dedication

This work is dedicated to our parents, whose support, guidance, and quiet encouragement have shaped our academic journey. Their patience, values, and constant motivation have been a source of strength for us, and we remain sincerely grateful for everything they have done to help us reach this stage.

Abstract

This project focuses on developing a deep learning based image denoising system using Generative Adversarial Networks (GANs). Images captured in low light or uncontrolled environments often contain significant noise, which reduces clarity and affects the performance of downstream tasks such as recognition and analysis. To address this problem, the proposed model employs a U-Net generator paired with a PatchGAN discriminator, allowing it to restore cleaner and more visually consistent images from noisy inputs. The training framework combines multiple loss functions including L1 loss, perceptual loss, SSIM-based loss, and adversarial loss to improve both pixel-level accuracy and perceptual quality. The system is trained and evaluated using the CelebA dataset, with performance measured through PSNR and SSIM to assess reconstruction quality. The denoising module developed in this phase will serve as the foundation for the next stage of the project, which will extend the approach to multi-angle view generation from a single image. The results demonstrate that GAN-based denoising can effectively recover structure, texture, and fine details, making it a suitable component for more advanced image synthesis tasks in subsequent phases of the project.

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List of Abbreviations and Acronyms

Abbreviation	Full Form
GAN	Generative Adversarial Network
cGAN	Conditional Generative Adversarial Network
CNN	Convolutional Neural Network
U-Net	U-shaped Convolutional Neural Network
PSNR	Peak Signal-to-Noise Ratio
SSIM	Structural Similarity Index Measure
VGG	Visual Geometry Group Network
L1 Loss	Mean Absolute Error Loss
MSE	Mean Squared Error
AMP	Automatic Mixed Precision
LR	Learning Rate
GPU	Graphics Processing Unit
GT	Ground Truth

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Chapter 1

Introduction

1.1 Introduction

Image quality plays a central role in modern computer vision applications. However, images captured in uncontrolled or low light environments frequently suffer from significant noise, distortion, and loss of structural detail. Such degradation directly affects the accuracy, reliability, and effectiveness of computer vision systems. Traditional filtering methods can suppress noise, but they often oversmooth important textures and edges. This project focuses on building a GAN-based image denoising framework that restores noise corrupted images by learning a mapping between noisy and clean representations. A U-Net generator architecture is used to reconstruct finer details, while a PatchGAN discriminator guides the generator toward producing sharper and more natural results. The system incorporates multiple loss functions including L1 loss, perceptual loss using a VGG network, SSIM-based loss, and adversarial loss to ensure that both pixel-level accuracy and structural similarity are preserved. The denoising module completed in this phase will serve as the foundation for the second part of the project, which aims to generate multi-angle views of a face from a single cleaned input image. This extended capability will support applications in identity reconstruction, augmented reality, and facial analysis. This chapter provides an overview of the problem, background, motivations, and contributions of the research.

1.2 Problem Statement

Images degraded by noise pose several challenges in computer vision workflows:

- Loss of fine details and structural similarity.
- Reduced performance in recognition or detection systems.

- Difficulty in reconstructing accurate textures and facial features.
- Traditional denoising filters blur essential information.
- Models trained on clean data perform poorly on noisy inputs.

The primary problem addressed in this research is the restoration of high-quality, structurally accurate images from noisy inputs using a generative model capable of preserving facial features, textures, and identity information. This denoising stage must be reliable, as the next phase of the project depends on generating multi-angle views from the restored images.

1.3 Problem Background

Noise may arise due to sensor limitations, low illumination, environmental disturbances, or transmission errors. In many real-world applications such as surveillance and facial analysis, noisy images significantly reduce recognition accuracy. Classical filtering-based approaches (Gaussian blur, median filter, bilateral filter) apply fixed rules and often remove high-frequency information such as edges and textures. Deep learning models, particularly convolutional networks, have demonstrated superior performance in image restoration tasks. However, CNNs alone often produce smooth outputs that lack perceptual sharpness. Generative Adversarial Networks (GANs) improve on this by incorporating a discriminator that penalizes unrealistic details, encouraging the generator to produce more natural images. The U-Net architecture is especially effective in image-to-image translation tasks due to its skip connections, which allow low-level features to be preserved during reconstruction. Combining U-Net with a GAN framework provides a powerful solution for removing noise while maintaining structural consistency.

1.4 Research Objectives

The objectives of this research are categorized into conceptual and technical goals.

1.4.1 Conceptual Objectives

- To examine the limitations of classical and CNN-based denoising methods.
- To analyze GAN-based approaches for perceptual image restoration.
- To understand the role of multi-loss optimization in improving reconstruction.

1.4.2 Technical Objectives

- To design and implement a GAN-based image denoising model using a U-Net generator and PatchGAN discriminator.
- To integrate L1, perceptual, SSIM, and adversarial losses for improved visual quality.
- To train the model using the CelebA dataset and evaluate performance using PSNR and SSIM.
- To use Automatic Mixed Precision (AMP) for faster and more stable training.
- To prepare the denoising module for future extension into multi-angle view generation.

1.5 Motivations

The increasing dependence on digital images for recognition, verification, and reconstruction motivates the need for clean, high-quality visual data. Noisy images significantly impair downstream tasks, leading to inaccurate results and unreliable decisions. GAN-based denoising techniques offer a promising solution by preserving texture, edges, and structural details that traditional methods struggle to maintain. This research is also motivated by the long-term goal of integrating denoising with multi-angle view generation. Cleaned images are essential for accurate pose synthesis, 3D reconstruction, and identity-preserving transformations.

1.6 Significance of the Research

This research holds significance for multiple reasons:

- It provides an effective deep learning method for restoring noisy facial images.
- The multi-loss optimization strategy enhances both structural similarity and perceptual sharpness.
- The model enables improved downstream performance in recognition and synthesis tasks.
- It establishes a strong foundation for multi-angle view generation in the next project phase.

The resulting model is useful for applications involving low-light imaging, surveillance, identity preservation, and augmented reality.

1.7 Research Contribution

This work contributes the following:

- A complete GAN-based image denoising framework integrating U-Net and PatchGAN.
- A loss function design combining L1, perceptual, SSIM, and adversarial losses.
- An optimized training pipeline using Automatic Mixed Precision.
- Evaluation of results using PSNR and SSIM, demonstrating improved reconstruction quality.
- Preparation of a denoising module compatible with multi-angle image generation.

1.8 Complex Engineering Analysis

1.8.1 Complex Engineering Problems

The project involves solving problems related to high-dimensional data, nonlinear feature extraction, noise modeling, and adversarial training stability. GAN training is inherently complex because it requires balancing two competing networks, managing mode collapse, and tuning sensitive hyperparameters.

1.8.2 Complex Engineering Activities

The work requires designing deep architectures, implementing multi-loss optimization, managing GPU computation, processing large datasets, and evaluating structural metrics. These tasks require advanced engineering judgment and systematic problem-solving, meeting the criteria of complex engineering activities.

1.9 Thesis Organization

This thesis is organized into six chapters to present the research in a structured and coherent manner. Chapter 1 introduces the background of the study, outlines the problem, and highlights the motivations, objectives, and contributions. Chapter 2 provides the necessary background study and reviews related literature on image denoising and GAN-based models. Chapter 3 explains the research methodology, including the proposed framework, dataset analysis, and model architecture. Chapter 4 presents the implementation details, evaluation metrics, experimental results, and discussions. Chapter 5 outlines the relevant standards, constraints, ethical considerations, and project milestones. Finally, Chapter 6 concludes the work by summarizing the findings, identifying limitations, and suggesting future research directions.

Chapter 2

Background Study

2.1 Introduction

Image restoration has become an essential part of computer vision as noisy or low-quality images negatively affect recognition accuracy, identity verification, and high-level visual understanding. In particular, face images captured under poor lighting, sensor limitations, or environmental disturbances often contain significant noise. This chapter provides the necessary background concepts related to image denoising, deep learning-based restoration, and Generative Adversarial Networks (GANs), which form the basis of the proposed denoising framework.

2.2 Literature Review

Early image denoising techniques such as Gaussian smoothing, median filtering, and bilateral filtering reduce noise by smoothing pixel intensities. Although effective for basic noise removal, these techniques tend to oversmooth the image and destroy important details such as edges and facial textures.

Patch-based classical methods like Non-Local Means (NLM) and BM3D advanced the field by comparing similar patches across an image and collaboratively filtering them. Despite improved performance, they still face limitations when dealing with complex face structures or high noise levels.

The emergence of deep learning revolutionized image restoration. Convolutional neural network (CNN)-based denoising models such as DnCNN demonstrated that learning noise patterns directly from data significantly improves reconstruction quality. However, CNNs optimized with pixel-wise losses often produce blurry or over-smoothed outputs.

Generative Adversarial Networks (GANs), introduced by Goodfellow et al., pro-

vided a new direction by enabling models to learn distributions of clean images rather than applying fixed transformations. GAN-based methods generate visually realistic, sharper, and more detailed denoised images by leveraging adversarial learning. The discriminator enforces naturalness by penalizing unrealistic textures, while the generator learns to produce outputs indistinguishable from real clean images.

U-Net architectures have become widely used in restoration tasks due to their encoder-decoder structure and skip connections, which preserve spatial details during reconstruction. PatchGAN discriminators evaluate images at the patch level, improving the preservation of fine textures and local details. Additionally, perceptual loss and SSIM-based loss enhance structural consistency and ensure that restored images maintain identity characteristics.

These developments collectively motivate the design of a GAN-based denoising system capable of restoring facial images with high perceptual quality and structural accuracy.

2.3 Problem Analysis

Noisy facial images present several challenges that make restoration difficult. Noise corrupts critical facial details such as skin texture, edges, and subtle identity cues. Since faces contain delicate features, even small distortions can degrade recognition performance.

Classical filtering approaches fail because they cannot distinguish between noise and fine facial details—they treat both as high-frequency components and remove them indiscriminately. CNN-based models improve structural recovery but still struggle with perceptual sharpness when trained with simple pixel-level losses like MSE or L1.

GAN-based restoration is more suitable for this problem because:

- The generator learns complex mappings from noisy to clean images.
- The discriminator enforces realism and texture-level accuracy.
- Perceptual and SSIM losses preserve identity-related details.

- U-Net connections allow texture information to be retained across layers.

The CelebA dataset adds further complexity due to variations in lighting, pose, background, and facial structures. A denoising model must generalize across these variations while producing consistent results. Evaluating performance using PSNR and SSIM helps quantify both pixel similarity and structural similarity, which are essential for reliable face restoration.

Overall, the problem requires a model capable of restoring clean images without losing important facial features, balancing pixel accuracy with perceptual realism—challenges that a GAN-based framework is well positioned to address.

Chapter 3

Research Methodology

3.1 Introduction

This chapter presents the methodology followed to design, train, and evaluate the proposed GAN-based image denoising system. It explains the overall workflow, dataset preparation, model architecture, training strategy, and modular structure of the system. The primary objective of this methodology is to restore noisy facial images with high perceptual quality and structural accuracy. The denoising module also serves as the foundation for the second phase of the project, which aims to generate multi-angle facial views from clean images.

3.2 Proposed Framework

The proposed system follows an image-to-image translation framework, where a noisy facial image is mapped to a clean reconstructed version using a Generative Adversarial Network (GAN). A U-Net generator is used to reconstruct image details, and a PatchGAN discriminator is used to evaluate the realism of the reconstructed output. Multiple loss functions L1, perceptual, SSIM, and adversarial jointly guide the learning process.

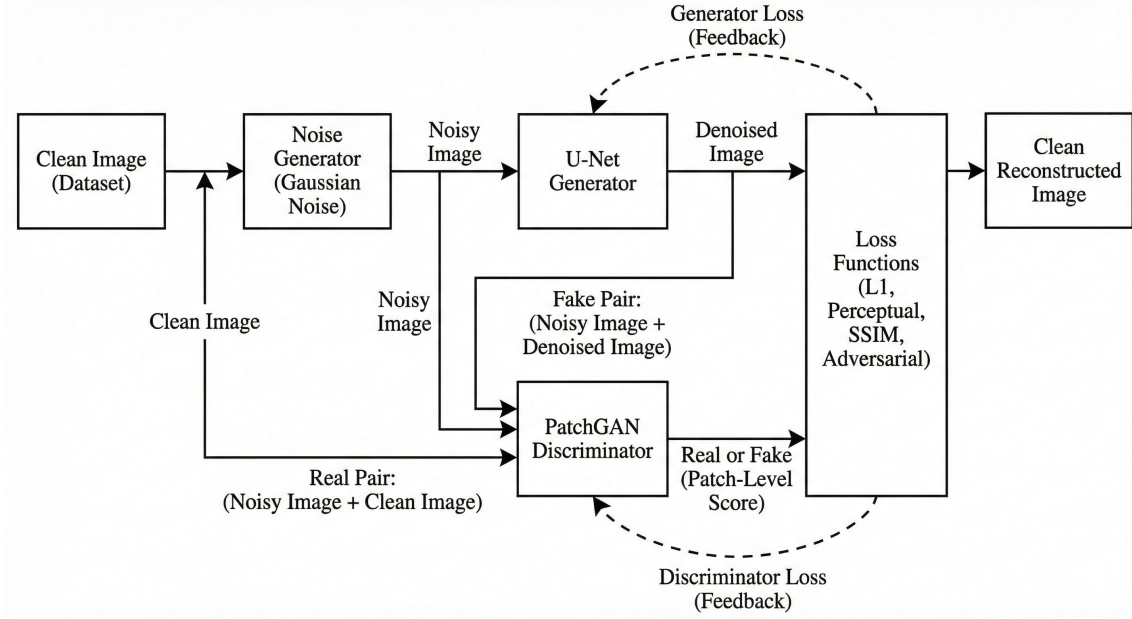


Figure 3.1: Workflow Diagram of the GAN-based Image Denoising System

The workflow begins with clean images from the dataset, which are corrupted using Gaussian noise to generate paired noisy samples. The noisy image is passed through the U-Net generator to produce a denoised result. The PatchGAN discriminator receives both real (noisy + clean) and fake (noisy + generated) pairs to classify them at the patch level. Feedback from the discriminator and the multi-loss functions jointly optimize the generator to produce clean reconstructed images.

3.3 Data Set Analysis

The CelebA dataset was used for training and evaluating the denoising model. It contains over 200,000 aligned human face images with variations in pose, lighting, background, and expression. A subset of 2,400 images was selected for this project to maintain manageable training time while retaining diversity.

3.3.1 Dataset Description

CelebA images were resized to 128×128 pixels and converted into tensors. Pixel values were normalized to the range $[0,1]$. Gaussian noise was added to each clean image to generate noisy-clean pairs for supervised learning.

3.3.2 Data Collection and Preprocessing

To ensure consistency and efficient training, the following preprocessing steps were applied:

- Resizing all images to a uniform resolution.
- Normalizing pixel values to stabilize training.
- Applying Gaussian noise to simulate realistic distortions.
- Creating mini-batches using PyTorch DataLoader.

These steps ensure that the generator learns a precise mapping from noisy images to their corresponding clean targets.

3.3.3 Feature Characteristics

CelebA includes a wide range of facial textures and lighting conditions, which helps the model generalize better. Variations in gender, skin tone, hairstyle, and illumination provide strong diversity for training deep learning models.

Total Images Used	2400 (CelebA Subset)
Image Resolution	128×128 pixels
Noise Type Applied	Gaussian Noise ($\sigma = 0.10$)
Train-Validation Split	2000 Train, 400 Validation

Table 3.1: Dataset Summary for GAN-based Image Denoising

3.4 Algorithm / Model Analysis

The denoising model is built using a U-Net generator and a PatchGAN discriminator. The generator restores fine details, while the discriminator evaluates realism in local image patches.

3.4.1 U-Net Generator

The U-Net generator consists of an encoder–decoder structure with skip connections. The encoder reduces spatial dimensions while learning feature representations, and the decoder reconstructs the image while reusing encoder features through skip paths. This design helps preserve facial textures and edges during denoising.

3.4.2 PatchGAN Discriminator

PatchGAN evaluates images at the patch level rather than processing the entire image at once. By classifying each patch as real or fake, the discriminator encourages the generator to produce sharp, realistic details across the entire image.

3.4.3 Loss Function Design

The model uses a combination of complementary loss functions:

- **L1 Loss:** Ensures pixel-level accuracy.
- **Perceptual Loss:** Uses VGG features to preserve high-level visual structure.
- **SSIM Loss:** Maintains structural similarity between predicted and ground truth images.
- **Adversarial Loss:** Encourages realistic texture reconstruction.

These losses jointly ensure that the output is sharp, clean, and structurally consistent.

3.4.4 Training Strategy

Training is conducted using:

- Adam optimizer with a learning rate of 1×10^{-4}
- Automatic Mixed Precision (AMP) for efficiency
- Alternating updates for generator and discriminator

The model is first warmed up with L1 loss before GAN loss is applied to stabilize training.

3.5 Explanation of Different Modules

The system consists of two major modules:

3.5.1 Module 1: GAN-based Image Denoising (Completed)

This module removes noise and produces clean facial images using the trained GAN. It includes noise generation, generator–discriminator training, multi-loss optimization, and quality evaluation using PSNR and SSIM.

3.5.2 Module 2: Multi-Angle View Generation (Future Work)

This module will use the denoised output to synthesize multiple facial angles using a generative model. It is part of the next phase of the research and depends on clean images produced from Module 1.

3.6 Summary

This chapter described the complete methodology used to develop the GAN-based image denoising system. It detailed the dataset analysis, model architecture, training strategy, and system modules. The workflow diagram provided a visual overview of the entire pipeline. The next chapter presents implementation details and experimental results.

References

- [1] I. Goodfellow et al., “Generative Adversarial Nets,” *Advances in Neural Information Processing Systems*, pp. 2672–2680, 2014.
- [2] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional Networks for Biomedical Image Segmentation,” *MICCAI*, pp. 234–241, 2015.
- [3] P. Isola, J. Zhu, T. Zhou, and A. Efros, “Image-to-Image Translation with Conditional Adversarial Networks,” *IEEE CVPR*, pp. 1125–1134, 2017.
- [4] K. Zhang et al., “Beyond a Gaussian Denoiser: Residual Learning of Deep CNN for Image Denoising,” *IEEE Transactions on Image Processing*, vol. 26, no. 7, pp. 3142–3155, 2017.
- [5] J. Johnson, A. Alahi, and L. Fei-Fei, “Perceptual Losses for Real-Time Style Transfer and Super-Resolution,” *ECCV*, pp. 694–711, 2016.
- [6] Z. Wang, A. Bovik, H. Sheikh, and E. Simoncelli, “Image Quality Assessment: From Error Visibility to Structural Similarity,” *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, 2004.
- [7] Z. Liu et al., “Deep Learning Face Attributes in the Wild,” *IEEE ICCV*, pp. 3730–3738, 2015. (CelebA Dataset)
- [8] C. Ledig et al., “Photo-Realistic Single Image Super-Resolution Using a GAN,” *IEEE CVPR*, pp. 4681–4690, 2017.
- [9] T. Karras, S. Laine, and T. Aila, “A Style-Based Generator Architecture for Generative Adversarial Networks,” *IEEE CVPR*, pp. 4401–4410, 2019.
- [10] M. Mirza and S. Osindero, “Conditional Generative Adversarial Nets,” *arXiv preprint*, arXiv:1411.1784, 2014.
- [11] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, “Unpaired Image-to-Image Translation using CycleGAN,” *IEEE ICCV*, pp. 2223–2232, 2017.
- [12] L. Rudin, S. Osher, and E. Fatemi, “Nonlinear Total Variation Based Noise

Removal Algorithms,” *Physica D*, vol. 60, pp. 259–268, 1992.

[13] W. Shi et al., “Real-Time Single Image and Video Super-Resolution Using an Efficient Sub-Pixel CNN,” *IEEE CVPR*, pp. 1874–1883, 2016.

[14] A. Dosovitskiy et al., “Generating Images with Perceptual Similarity Metrics based on Deep Networks,” *NIPS Workshops*, 2016.

[15] A. Tran, T. Hassner, I. Masi, and G. Medioni, “Reconstruction of 3D Faces from a Single Image via Deep Neural Network,” *IEEE CVPR Workshops*, pp. 1806–1815, 2017.